

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2016-17

Paper No. GLB-503: Sedimentary and Metamorphic Petrology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/174

*Instructions: Attempt any **five** questions, selecting at least **two** questions each from Section A and B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Discuss in detail the processes responsible for the formation of sedimentary rocks. [14]

Question 2. Describe the classification of conglomerates and characterize their genesis. [14]

Question 3. Write detailed notes on **any two** of the following: [2 × 7 = 14]

- (a) Argillaceous rocks
- (b) Cross-bedding
- (c) Nonclastic texture
- (d) Heavy minerals

SECTION — B

Question 4. Describe progressive metamorphism of ultramafic rocks in the $\text{MgO} - \text{Al}_2\text{O}_3 - \text{SiO}_2 - \text{H}_2\text{O}$ system. [14]

Question 5. Discuss the metamorphic facies and metamorphic facies series. [14]

Question 6. Write notes on **any two** of the following: [2 × 7 = 14]

- (a) Goldschmidt's mineralogical Phase Rule
- (b) Prograde Metamorphism
- (c) ACF and AKF diagrams

Question 7. Describe progressive metamorphism of Pelitic rocks in the $\text{K}_2\text{O} - \text{FeO} - \text{MgO} - \text{Al}_2\text{O}_3 - \text{SiO}_2$ system. [14]

SECTION — C (Compulsory)

Question 8. Fill in the blanks: [7 × 2 = 14]

- (a) Eclogites are composed of _____ and _____.
- (b) Quartzarenites are texturally _____ and mineralogically _____.
- (c) In the series Slate — Phyllite — Schist, the grain size of the rock _____.
- (d) Small-scale ripples have a height of _____.
- (e) Hornblende + quartz = _____ + diopside + plagioclase + H₂O.
- (f) Freshly deposited clay has a porosity of _____.
- (g) Corundum + quartz = _____ (Choose from: *sillimanite* / *sapphirine* / *cordierite*).